

Instructions

- Answer all eight questions. Show sufficient working for method marks.
- A calculator may be used unless a question states otherwise.
- Write clearly in the spaces provided. Additional pages may be attached.

1

Calculate the number of moles in 12.0 g of carbon-12, showing the relationship you use.

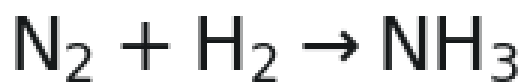
[4 marks]

$$n = \frac{m}{M}$$

2

Balance the equation and identify the limiting reactant when 4 mol of each reactant is supplied.

[5 marks]



3

Describe how you would determine the empirical formula of a compound from its percentage composition.

[6 marks]

4

Explain, in terms of collision theory, why increasing temperature increases the rate of reaction.

[5 marks]

5

Determine the concentration of a solution made by dissolving 5.85 g of NaCl in 250 cm³ of water.

[5 marks]

$$C = \frac{n}{V}$$

6

Calculate the percentage yield when 8.4 g of product is obtained from a theoretical maximum of 10.0 g.

[4 marks]

7

Explain why ionic compounds conduct electricity when molten but not when solid.

[5 marks]

8

Describe the trend in reactivity down Group 1 and explain it in terms of atomic structure.

[6 marks]
